

## Problem 5.1

$$a) \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$b) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} ??$$

## Problem 5.2

$$a) 10$$

$$b) 6$$

## Problem 5.3

$$a) \text{ True } (A+B)^T = A^T + B^T = A+B \text{ and } (cA)^T = cA$$

$$b) \text{ True } (A+B)^T = A^T + B^T = -(A+B) \text{ and } (cA)^T = cA$$

$$c) \text{ False } \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$